IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE/EIE PART III/IV: MEng, BEng and ACGI

BIOMEDICAL ELECTRONICS

Tuesday, 9 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR out of five questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : T. Constandinou

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1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from muscles.*

   a) Explain how a neural action potential is generated, referring to the ionic flow of sodium and potassium ions. As part of your answer, sketch the waveform of an action potential, annotating and explaining the mechanism contributing to each phase.

   [4]

   b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record from a set of electrodes connected to the back muscles shown in Figure 1.1.

   Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

   Instrumentation Amplifier Gain: 21
   Low-Pass Filter Roll off: 40dB/decade
   High-Pass Filter Roll off: 60dB/decade
   Supply Range: -5V to 5V

   State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters.

   [10]

   c) The function of the system is to measure the muscle fiber conduction velocity (MFCV), determined by the propagation delay between the signals measured between electrodes 1-2 and 3-4. Electrode 5 is used to set the common mode voltage of the body. Design a system using your front end designed in 1.b which is able to measure the MFCV and output a digital number representing this value. You are only required to sketch the top-level architecture. State all assumptions in your design.

   [6]

Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

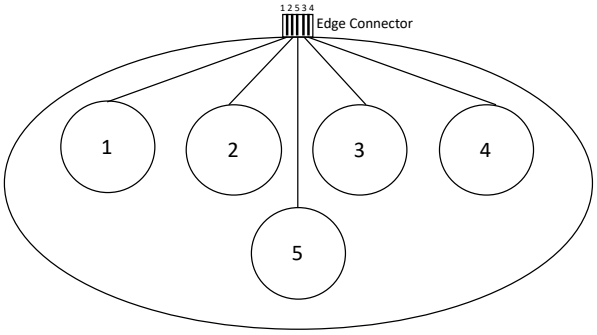


Figure 1.1

2. This question relates to electrical stimulation applied to bioelectronic medicines:

Bioelectronic medicines are implantable devices that modulate peripheral or autonomic nerves (such as the cervical vagus nerve) to alter end-organ function, yielding therapeutic benefits in systemic inflammatory disorders (e.g., rheumatoid arthritis, Crohn's disease).

a) List the main electrode types used for peripheral-nerve stimulation. For each, state one advantage and one limitation. [5]

b) The human cervical vagus nerve is approximately 5 mm in diameter. Sketch a cuff-based electrode whose geometry increases stimulation selectivity relative to a simple two-ring cuff and explain briefly why this design is more selective. [5]

c) Using the smallest active contact area proposed in your electrode design in part (b), assess the viability of the electrode being used to deliver a biphasic, cathodic-first stimulus of 1mA with a 800µs phase width. Assume platinum-iridium has a reversible charge-density limit of $200\mu Ccm^{-2}$ and that the fabrication process yields an interface impedance of $12\Omega cm^2$ at 1kHz.

(i) Estimate the maximum cathodic current the contact can safely deliver. [2]

(ii) Calculate the minimum voltage compliance the stimulator must provide. [3]

d) Based on the compliance calculated in part (c), outline the key requirements for the output stage of a current-mode neural stimulator. [5]

3. This question relates to the design of an instrument for detecting DNA.

a) Explain two methods which can be used to detect DNA with an electrochemical sensor, describing the physical phenomena which allow these detection methods and stating one advantage and one disadvantage for each method.

[6]

b) Figure 2.1 shows typical reaction measured from an ISFET sensor used for a bacterial test, where there is a change in pH caused by the DNA of the bacteria binding to its complementary target. The output voltage shown is linearly dependent on the pH change and the sensor has baseline drift. Figure 2.1a is a positive confirmation of the bacteria and Figure 2.1b is a negative control reaction to measure the baseline drift of the sensor. Given the data in Figure 2.1, design a suitable diagnostic system to measure bacterial DNA using ISFETs, sketching its system architecture and circuits for instrumentation.

[8]

c) The circuit shown in Figure 2.2 is an advanced ISFET front end which outputs a signal V_{out} in the form of a pulse whose width is proportional to pH. Its operation is based on two phases, a reset phase when RST = 0 and a readout phase where RST = V_{DD}. The ISFET sensor M_0 is operating in strong inversion.

i. Derive the relationship between pH and the length of the output voltage pulse V_{out}, stating any assumptions made.

[3]

ii. Show how the architecture in Figure 2.2 can be used to instrument to a 3x3 array of ISFET sensors.

[3]

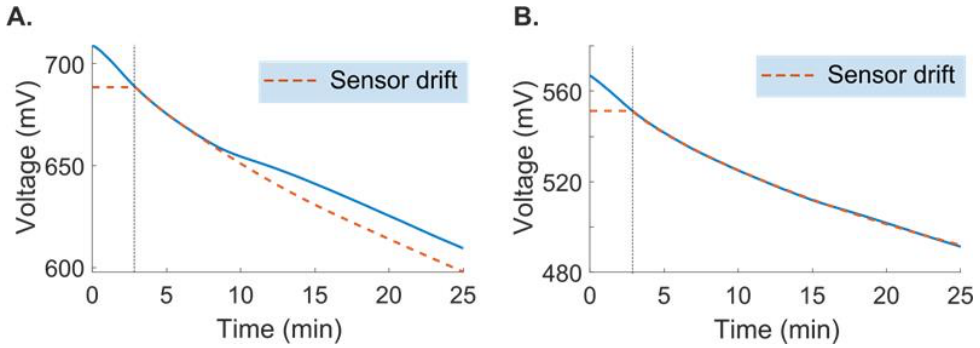


Figure 2.1

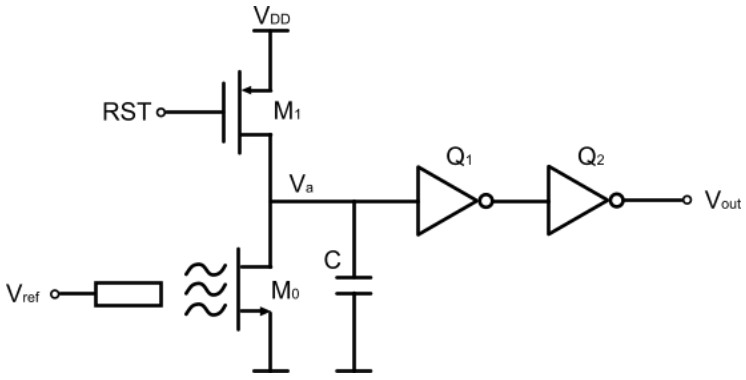


Figure 2.2

4. This question relates to a cochlear implant:

   a) Explain why most commercial cochlear implants typically use a monopolar stimulation configuration rather than bipolar. [3]

   b) A behind the ear (BTE) sound processor feeds a 22-contact electrode array via an implanted receiver-stimulator.

      (i) Sketch the basic architecture of the complete cochlear implant system. [3]

      (ii) List five key functional blocks between the microphones and the electrodes, and for each block state its principal design constraint (e.g. power, size, latency, EMC, safety, etc.) with a one-sentence justification. [5]

   c) A fully implantable research prototype uses a 3.7V, 15mAh Li-ion rechargeable micro-battery to power a custom chip that draws a steady 126µW. The battery is recharged periodically via the same 13.56MHz inductive link that carries calibration data. Calculate the maximum operating time, in hours, between charges if the 126µW demand is truly constant. State two reasons why the device is likely to consume more than stated, and one practical reason why the real-world operating time might be shorter than your calculated figure. [5]

   d) Name two clinically used hearing-restoration devices other than the cochlear implant. For each device briefly describe its working principle, state one clear advantage it offers relative to a conventional cochlear implant and identify one engineering or clinical challenge that has limited its wider adoption. [4]

5. This question relates to the design of an artificial pancreas for treatment of diabetes.

a)

i) Sketch a system level diagram of a closed loop system which works as an artificial pancreas and describe its operation.

[5]

ii) Explain three main disturbances to the system which would cause the blood glucose to fluctuate.

[3]

iii) Explain one challenge that is faced with current technology.

[2]

b)

The blood glucose of a diabetic is expected to fluctuate in the range of 0-20mM. A continuous glucose sensor consists of an enzyme which can generate 10nA of current for every 1mM of glucose when biased with a redox potential of 700mV. Design a complete instrumentation system to sense blood glucose using this enzyme which works off a 3V supply and maximizes the dynamic range available.

[10]